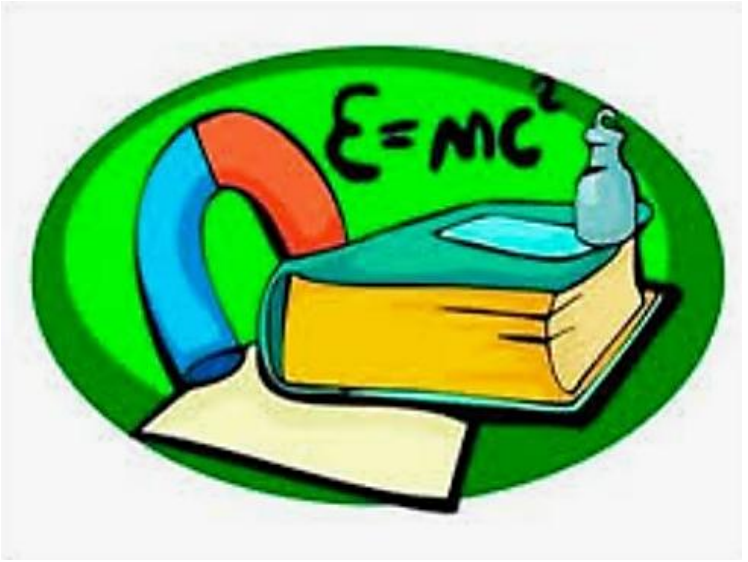
Fizika

9-sinf

Mavzu:

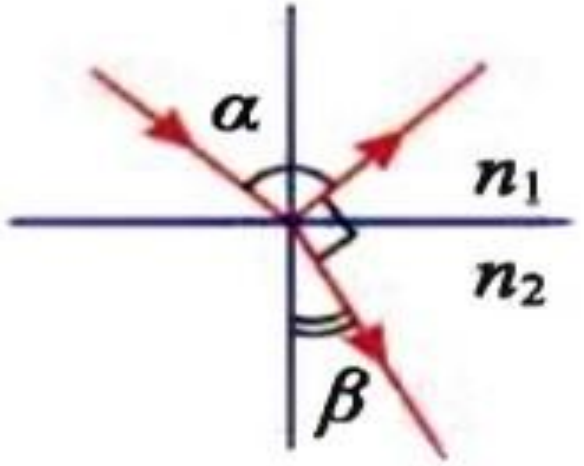
To‘la ichki qaytish



Yorug‘likning qaytish va sinish qonunlari

1. Tushgan nur, qaytgan nur va tushish nuqtasiga o‘tkazilgan perpendikulyar bir tekislikda yotadi.
2. Qaytish burchagi tushish burchagiga teng:

$$\alpha = \gamma$$

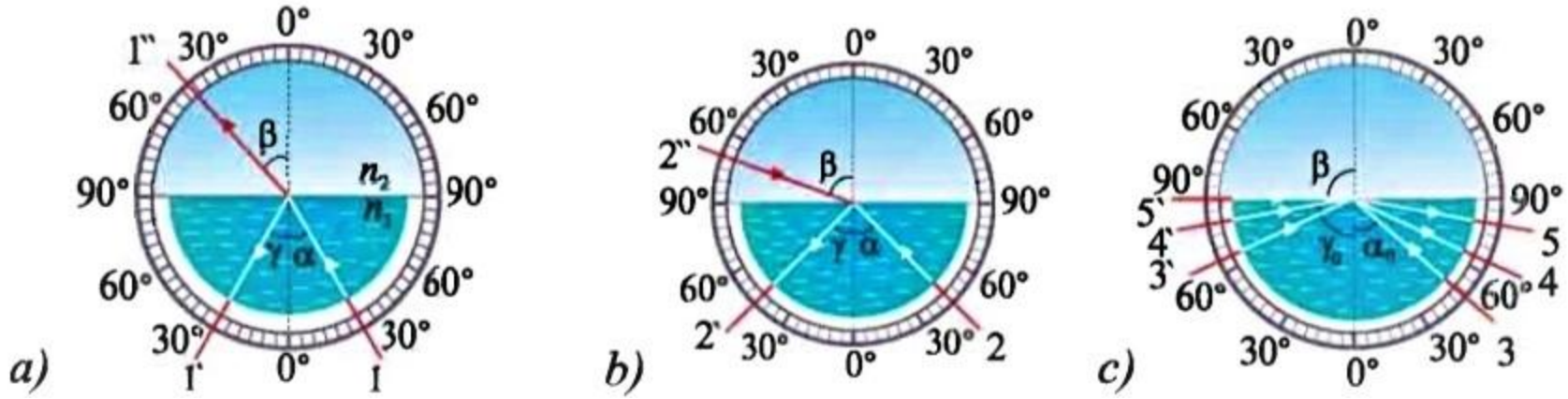


1. Tushgan nur, singan nur va ikki muhit chegarasiga nurning tushish nuqtasiga o‘tkazilgan perpendikulyar bir tekislikda yotadi.

2. Tushish burchagi sinusining sinish burchagi sinusiga nisbati ikki muhit uchun o‘zgarmas kattalikdir.

$$n = \frac{\sin \alpha}{\sin \beta}$$

Yorug‘likning to‘la qaytishi



Shisha uchun $\boldsymbol{n_1 = 1,5}$ va havo uchun $\boldsymbol{n_2 = 1}$ bo‘lsa,

$$\sin\beta = \frac{n_1}{n_2}\cdot\sin\alpha$$

$\alpha = 30^0$ da $\beta = 42^0$, $\alpha = 40^0$ da $\beta = 75^0$,

$\alpha = \alpha_0$ da $\beta = 90^0$

To‘la ichki qaytish

$$\sin\alpha_0 = \frac{n_2}{n_1} \quad , \quad \sin\alpha_0 = \frac{1}{1{,}5} \approx 0{,}667 \quad , \quad \alpha_0 \approx 42^{\circ}$$

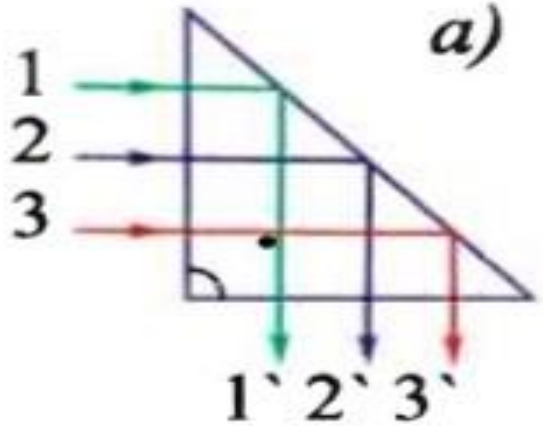
$$\boldsymbol{\sin\alpha_0 = \frac{1}{n}}$$

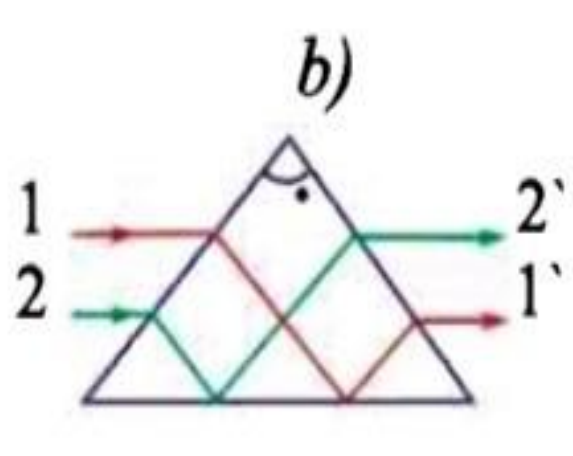
Sindirish ko‘rsatkichi katta bo‘lgan muhitdan sindirish ko‘rsatkichi kichik bo‘lgan muhitga yorug‘lik yo‘naltirilganda tushish burchagi ma’lum burchakdan katta bo‘lganda nur ikki muhit chegarasidan ***to‘la qaytadi***.

Bu ***to‘la ichki qaytish hodisasi*** deyiladi.

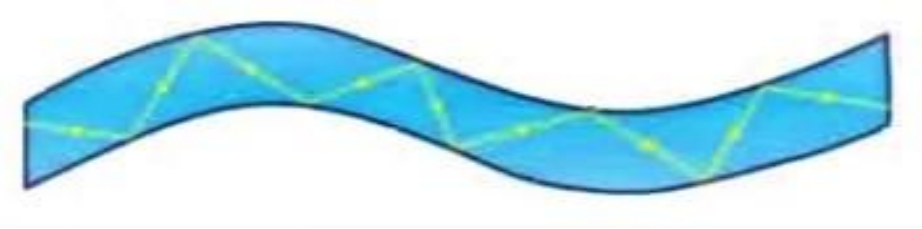
Qo‘llanilishi

To‘la ichki qaytishdan yorug‘lik nurlarini biror yo‘nalishda burish yoki nurlar dastasini o‘rnini almashtirish uchun foydalaniladi.





Nur tolalar optikasi



To‘la ichki qaytish hodisasidan axborot texnologiyasida, aloqa tizimida, tibbiyotda va boshqa sohalarda keng foydalaniladi.

Masala

Havoga nisbatan toza suv, shisha va olmosning nur sindirish koʻrsatkichlari **1,33; 1,5** va **2,42** ga teng. Shu moddalarning qaysi birida toʻla ichki qaytishning chegaraviy burchagi eng kichik?

Berilgan:

$1)\ n = 1{,}33$

$2)\ n = 1{,}5$

$3)\ n = 2{,}42$

Topish kerak: $\alpha_0 - ?$

Formula va yechish:

$$\sin\alpha_0 = \frac{1}{n}$$

$$\alpha_0 = \arcsin\frac{1}{n}$$

$$1)\ \alpha_0 = \arcsin\frac{1}{1{,}33} = 49^o$$

$$2)\ \alpha_0 = \arcsin\frac{1}{1{,}5} = 42^o$$

$$3)\ \alpha_0 = \arcsin\frac{1}{2{,}42} = 24^o$$

Javob: olmosda to‘la ichki qaytishning chegaraviy burchagi eng kichik.

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(141-bet)
2. 26-mashqning 1-masalasini yechish.